

tSNR in ROIs

Libraries and Functions

```
# Clear environment
rm(list = ls())

# Get Libraries and Functions
source("~/Dropbox (Personal)/TI_paper/ti-paper/Data/Functions.R")
```

Import data

```
setwd("~/Dropbox (Personal)/TI_paper/ti-paper/Data/MRI/FaceNameTask/tSNR/")
ROIdata <- read.csv("tSNR.csv")

# Assign factors
ROIdata$S_ID <- factor(ROIdata$S_ID)
ROIdata$ID <- factor(ROIdata$ID)
ROIdata$ROI <- factor(ROIdata$ROI)
ROIdata$hemisphere <- factor(ROIdata$hemisphere)
```

Stats and plot

```
tSNR.m <- lmer(tSNR ~ hemisphere + (1|S_ID) + (1|ROI), data = ROIdata, REML = TRUE)
Anova(tSNR.m, test="F")
```

```
## Analysis of Deviance Table (Type II Wald F tests with Kenward-Roger df)
##
## Response: tSNR
##           F Df Df.res    Pr(>F)
## hemisphere 13.528  1    77 0.0004331 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(tSNR.m, ddf="Kenward-Roger")
```

```
## Linear mixed model fit by REML. t-tests use Kenward-Roger's method [
## lmerModLmerTest]
## Formula: tSNR ~ hemisphere + (1 | S_ID) + (1 | ROI)
##   Data: ROIdata
##
```

```
## REML criterion at convergence: 962.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.32204 -0.39978  0.03308  0.55743  2.71288
##
## Random effects:
##   Groups   Name                Variance Std.Dev.
##   S_ID      (Intercept)    938.9    30.64
##   ROI       (Intercept)    449.3    21.20
##   Residual                   1064.9    32.63
## Number of obs: 96, groups:  S_ID, 16; ROI, 3
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   192.098     15.187    4.011  12.649 0.000221 ***
## hemisphereR  -24.501      6.661   77.000   -3.678 0.000433 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## hemisphereR -0.219
```

```
tSNR.contrasts <- emmeans(tSNR.m, specs = pairwise ~ hemisphere, type = "response")
tSNR.contrasts
```

```
## $emmeans
## hemisphere emmean   SE    df lower.CL upper.CL
## L           192 15.2 4.01     150      234
## R           168 15.2 4.01     125      210
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate   SE df t.ratio p.value
## L - R           24.5 6.66 77   3.678  0.0004
##
## Degrees-of-freedom method: kenward-roger
```

```
p.tSNR.cortical <- ggplot(ROIdata, aes(x = hemisphere, y = tSNR, fill = hemisphere)) +
  geom_boxplot(alpha=1, outlier.shape = NA, notch = TRUE, position=position_dodge(0.8)) +
  scale_y_continuous(name = "tSNR",
                     breaks = seq(50, 300, 50),
                     limits=c(50, 300)) +
  scale_x_discrete(name = "Hemisphere") +
  theme(plot.title = element_blank(),
        axis.title.x = element_text(size = 14, family = "Arial", face = "bold"),
        axis.title.y = element_text(color="black", size=16, face="bold"),
        axis.text.x=element_text(color="black", size=16, face="bold"),
        axis.text.y = element_text(size = 14),
        legend.background = element_blank(),
```

```

    legend.key=element_blank(),
    panel.background = element_blank(),
    legend.title=element_blank()) +
  scale_colour_manual(values=c("#39cc4d", "#42ecf5"))+
  scale_fill_manual(values=c("#39cc4d", "#42ecf5"),
    name = "Hemisphere",
    breaks = c("L", "R"),
    labels = c("LH", "RH")) +
  theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
    strip.text.y = element_text(size = 20), legend.position = "none") + ggtitle("")

ggsave(p.tSNR.cortical, filename = "tSNR_cortex.tiff", width = 7, height = 9, dpi = 300, units = "cm", c

```